

Question #1:

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    string str;
```

```
    cout << "Enter a string: ";
```

```
    getline(cin, str);
```

```
    int length = 0;
```

```
    for (char c : str) {
```

```
        length++;
```

```
    }
```

```
    cout << "Length of the string: " << length << endl;
```

```
    ofstream outFile("chars.txt");
```

```
    outFile << str;
```

```
    outFile.close();
```

```
    ifstream inFile("chars.txt");
```

```
    char ch;
```

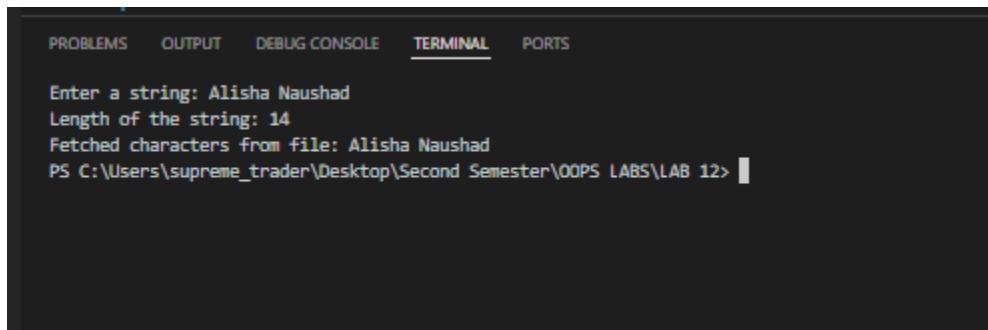
```
    cout << "Fetched characters from file: ";
```

```
    while (inFile.get(ch)) {
```

```
        cout << ch;
    }
    cout << endl;
    inFile.close();

    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Enter a string: Alisha Naushad
Length of the string: 14
Fetched characters from file: Alisha Naushad
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 12> |
```

Question #2:

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {
```

```
    ofstream makeSource("source.txt");
```

```
    makeSource << "Hello! This text is copied from source to destination.";
```

```
makeSource.close();

ifstream source("source.txt");
ofstream destination("destination.txt");

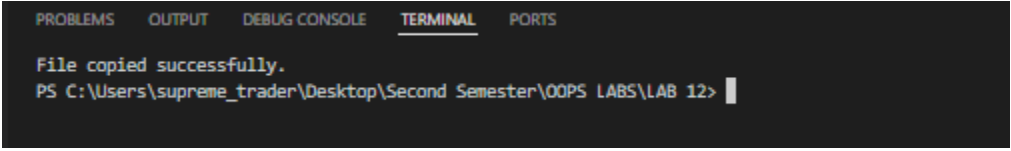
if (!source) {
    cout << "Error opening source file!" << endl;
    return 1;
}

char ch;
while (source.get(ch)) {
    destination.put(ch);
}

cout << "File copied successfully." << endl;

source.close();
destination.close();
return 0;
}
```

Output:



The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'TERMINAL' tab is selected. The output text in the terminal is 'File copied successfully.' followed by a new line and the command prompt 'PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 12>' with a cursor.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
File copied successfully.
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 12>
```

Question #3:

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <string>
```

```
#include <cstring>
```

```
using namespace std;
```

```
class Person {
```

```
public:
```

```
    char name[50];
```

```
    int age;
```

```
    Person() {}
```

```
    Person(string n, int a) {
```

```
        strcpy(name, n.c_str());
```

```
        age = a;
```

```
    }
```

```
};
```

```
int main() {
```

```
    Person person1("Alice", 25);
```

```
    ofstream outFile("person.bin", ios::binary);
```

```
    outFile.write((char*)&person1, sizeof(person1));
```

```
    outFile.close();
```

```
Person person2;

ifstream inFile("person.bin", ios::binary);

inFile.read((char*)&person2, sizeof(person2));

inFile.close();

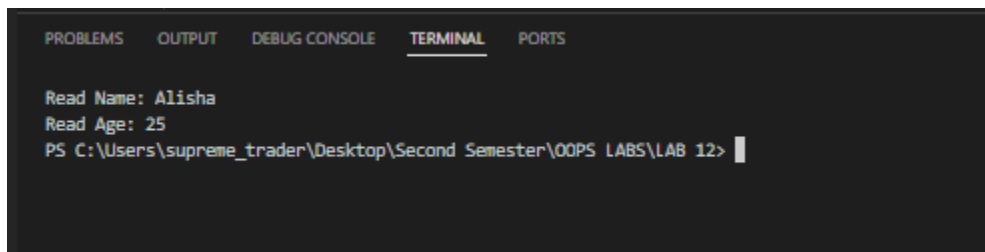

cout << "Read Name: " << person2.name << endl;

cout << "Read Age: " << person2.age << endl;


return 0;

}
```

Output:

A screenshot of a C++ IDE's terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (which is selected), and PORTS. The output text in the terminal is: "Read Name: Alisha", "Read Age: 25", and the command prompt "PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 12>".

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Read Name: Alisha
Read Age: 25
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 12>
```

Question #4:

```
#include <iostream>

#include <fstream>

#include <string>

#include <cstring>
```

```
using namespace std;
```

```
class Participant {
```

```
public:
```

```
    int id;
```

```
    char name[50];
```

```
    int score;
```

```
    void Input() {
```

```
        cout << "Enter ID: ";
```

```
        cin >> id;
```

```
        cout << "Enter Name: ";
```

```
        cin >> name;
```

```
        cout << "Enter Score: ";
```

```
        cin >> score;
```

```
        ofstream outFile("participant.dat", ios::binary | ios::app);
```

```
        if (outFile) {
```

```
            outFile.write(reinterpret_cast<char*>(this), sizeof(Participant));
```

```
            outFile.close();
```

```
        }
```

```
    }
```

```
    void Output() {
```

```
        int searchId;
```

```
        cout << "Enter ID to search: ";
```

```
cin >> searchId;

ifstream inFile("participant.dat", ios::binary);
if (!inFile) {
    cout << "No data file found." << endl;
    return;
}

Participant p;
bool found = false;

while (inFile.read(reinterpret_cast<char*>(&p), sizeof(Participant))) {
    if (p.id == searchId) {
        cout << "ID: " << p.id << "\nName: " << p.name << "\nScore: " << p.score << endl;
        found = true;
        break;
    }
}

inFile.close();

if (!found) {
    cout << "Participant not found." << endl;
}

void Max() {
```

```
ifstream inFile("participant.dat", ios::binary);  
if (!inFile) {  
    cout << "No data file found." << endl;  
    return;  
}
```

```
Participant p;
```

```
Participant maxP;
```

```
int maxScore = -1;
```

```
bool empty = true;
```

```
while (inFile.read(reinterpret_cast<char*>(&p), sizeof(Participant))) {
```

```
    empty = false;
```

```
    if (p.score > maxScore) {
```

```
        maxScore = p.score;
```

```
        maxP = p;
```

```
    }
```

```
}
```

```
inFile.close();
```

```
if (!empty) {
```

```
    cout << "Highest Score Participant:\nID: " << maxP.id << "\nName: " << maxP.name <<  
    "\nScore: " << maxP.score << endl;
```

```
    } else {
```

```
        cout << "No data in file." << endl;
```

```
}
```



```
    }  
};  
  
int main() {  
    Participant p;  
    int choice;  
  
    while (true) {  
        cout << "\n1. Input Data\n2. Output Data by ID\n3. Show Max Score\n4. Exit\nChoice: ";  
        if (!(cin >> choice)) {  
            cin.clear();  
            cin.ignore(10000, '\n');  
            cout << "Invalid choice. Please enter a number." << endl;  
            continue;  
        }  
  
        if (choice == 1) p.Input();  
        else if (choice == 2) p.Output();  
        else if (choice == 3) p.Max();  
        else if (choice == 4) break;  
        else cout << "Invalid choice." << endl;  
    }  
  
    return 0;  
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

1. Input Data
2. Output Data by ID
3. Show Max Score
4. Exit
Choice: 1
Enter ID: 345
Enter Name: Alisha
Enter Score: 67

1. Input Data
2. Output Data by ID
3. Show Max Score
4. Exit
Choice: 3
Highest Score Participant:
ID: 123
Name: Alisha
Score: 987

1. Input Data
2. Output Data by ID
3. Show Max Score
4. Exit
Choice: 345
Invalid choice.

1. Input Data
2. Output Data by ID
3. Show Max Score
4. Exit
Choice: 1
Enter ID: 23
Enter Name: sara
Enter Score: 45

1. Input Data
2. Output Data by ID
3. Show Max Score
4. Exit
Choice: 2
Enter ID to search: 34
Participant not found.

1. Input Data
2. Output Data by ID
3. Show Max Score
4. Exit
Choice: 2
Enter ID to search: 23
ID: 23
Name: sara
Score: 45

1. Input Data
2. Output Data by ID
3. Show Max Score
4. Exit
Choice: 3
Highest Score Participant:
ID: 123
Name: Alisha
Score: 987

1. Input Data
2. Output Data by ID
3. Show Max Score
4. Exit
Choice: 4
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 12> |
```

Question #5:

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <string>
```

```
using namespace std;
```

```
void countLines() {
```

```
    ifstream inFile("STORY.TXT");
```

```
    string line;
```

```
    int count = 0;
```

```
    if (!inFile) {
```

```
        cout << "Error opening file!" << endl;
```

```
        return;
```

```
    }
```

```
    while (getline(inFile, line)) {
```

```
        if (!line.empty() && line[0] != 'A') {
```

```
            count++;
```

```
        }
```

```
    }
```

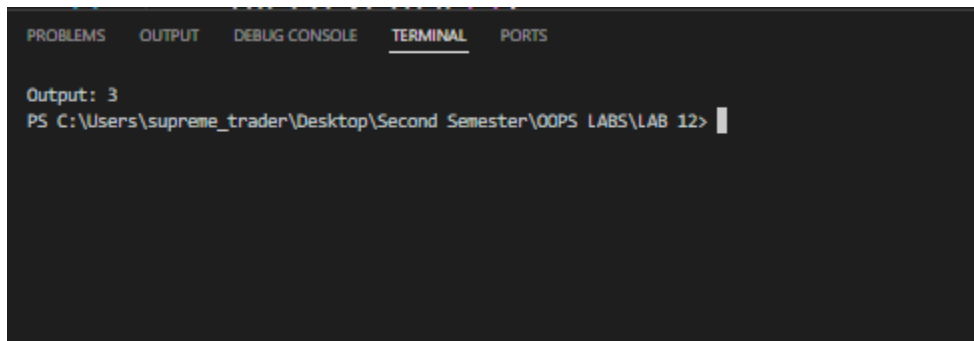
```
    inFile.close();
```

```
    cout << "Output: " << count << endl;
```

```
}
```

```
int main() {  
    ofstream outFile("STORY.TXT");  
  
    outFile << "The rose is red.\nA girl is playing there.\nThere is a playground.\nAn airplane is  
in the sky.\nNumbers are not allowed in the password.\n";  
  
    outFile.close();  
  
    countLines();  
}
```

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. Below the tabs, the text 'Output: 3' is displayed. The command prompt shows 'PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 12>' followed by a cursor.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
Output: 3  
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 12> |
```